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CMPE2100 - Hardware Interfacing (A02.1252), Eunice De Grace Fmukam Ngadjou, 4/14/26 at 6:15:38 AM MDT

Question1: *Score 18.75/30*

Project #8 In-Class -- Thermocouple Temperature and Control with the NI-USB6221

Note: Do not move on to the "Next" question until you have completed this In-Class section. The next question is the Build, Calibrate, and Test portion of the project, and once you move to that page, you will not be able to return to this page!

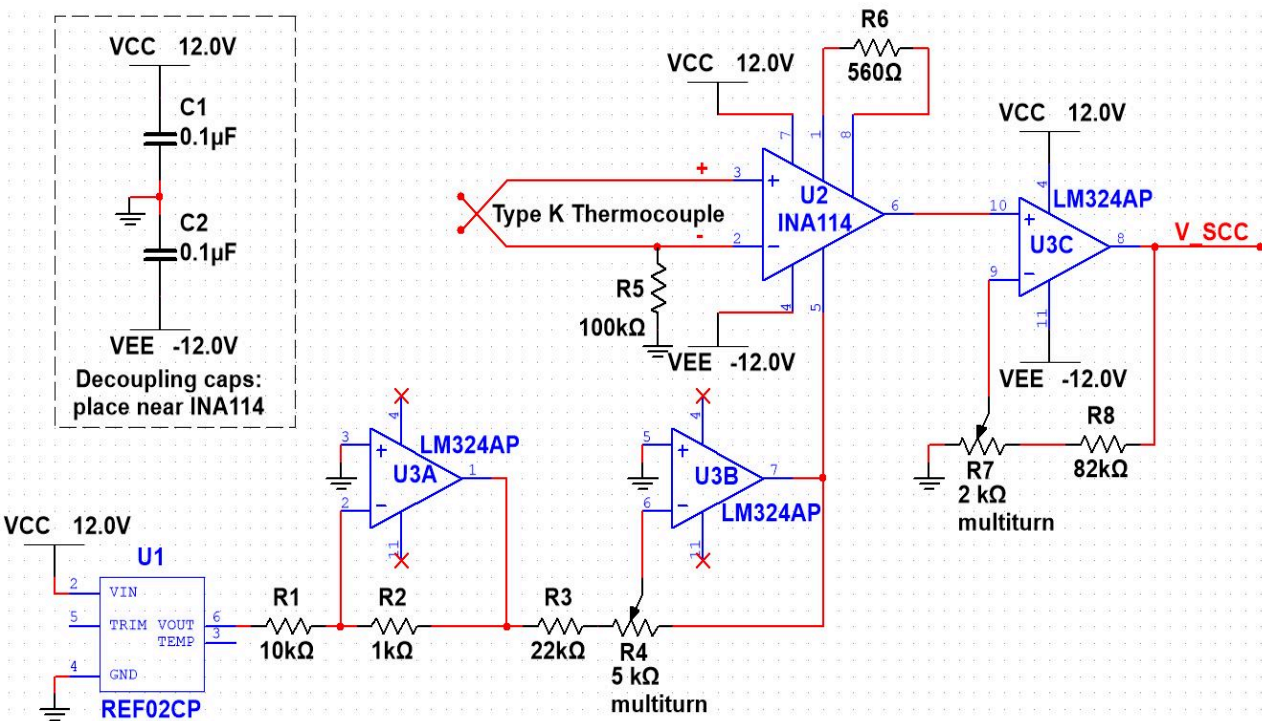
You are provided with one "How did I do?" to allow you to check your work upon completion of this question. At the end of the time allotted for the In-Class section, please "Save and Close" this question and move on to the next question -- do not "Submit" your work, as that would prevent you gaining access to the rest of the project.

Project Breakdown:

- 30 points -- In-Class Preparation
- 25 points -- Temperature Measurement SCC
- 30 points -- Human Interface
- 15 points -- Control Circuit

This activity, at first glance, looks quite similar to other activities you've recently completed. However, to provide you with broader experience with both electronics and LabVIEW, some hopefully interesting variations have been introduced. The entire circuit will be provided for you; however, you are expected to analyze the various parts of the circuit to demonstrate your ability to use basic electrical and electronic principles in configurations that are likely new to you.

Temperature Measurement Signal Conditioning Circuit



1. This circuit is intended to measure temperatures ranging from -15°C to +35°C. Using the [NIST table for the Type K thermocouple](https://srdata.nist.gov/its90/download/type_k.tab) (https://srdata.nist.gov/its90/download/type_k.tab), what are the minimum and maximum voltages expected from the measurement junction, in millivolts?

Your response	Correct response
-0.586	-0.586±0.002

Auto graded Grade: 1/1.0

mV to	
Your response	Correct response
1.407	1.407±0.001

Auto graded Grade: 1/1.0 ✓

mV.	
• The reference junction is assumed to be at 22°C. What is the expected range of differential voltages that will be measured by the instrumentation amplifier?	
Your response	Correct response
-1.465	-1.465±0.002

Auto graded Grade: 1/1.0 ✓

mV to	
Your response	Correct response
0.528	0.528±0.002

Auto graded Grade: 1/1.0 ✓

mV.	
• The output will be sent to the NI-USB6221 digitizer. To allow some room to work within the limitations of the amplifiers, the circuit output range will be limited to between -7.50 V and +7.50 V. How much gain is required for this circuit? (Don't be too surprised -- it's huge!)	
Your response	Correct response
7500	7,526±20

Auto graded Grade: 0/1.0 ✗

• Notice that the gain of the INA114 is not adjustable in this circuit. What is the expected gain of U2?	
Your response	Correct response
90.3	90.29±0.01

Auto graded Grade: 1/1.0 ✓

• The remaining gain must be handled by U3C. How much gain is required for this stage?	
Your response	Correct response
83.3	83.36±0.2

Auto graded Grade: 1/1.0 ✓

• Notice the unusual arrangement for setting the gain of U3C. This configuration saves one resistor compared to previous designs, but requires a bit of thought in the design phase, and a bit of care during calibration.	
1. Design considerations: The potentiometer is being used as a variable voltage divider instead of as a variable resistor as in other designs in this course. The part of it to the left of the sweeper is "R _i ", and the part of it to the right of the sweeper is part of "R _f ". Since this is a non-inverting amplifier, its gain is $A_v = \frac{R_f}{R_i} + 1$. However, R _i varies with adjustment -- as does R _f , which is the combination of the fixed resistor and the rest of the potentiometer. In terms of the reference designators given, $A_v = \frac{R8 + (R7 - R_i)}{R_i} + 1$. If R7 is all the way to the right (i.e. R _i = R7), what is this op amp's gain?	
Your response	Correct response
4	42

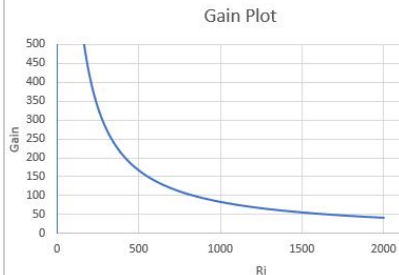
Auto graded Grade: 0/1.0 ✖

If R7 is all the way to the left, what can be said about the gain?

Your response	Correct response
it is practically infinite	it is practically infinite

Auto graded Grade: 1/1.0 ✔

A plot of the gains in response to changing R7 is shown below, truncated at a gain of 500 for clarity:



Notice that, from around $1000\ \Omega$ to $2000\ \Omega$, the change in gain is relatively moderate. So, if the R_i part of the potentiometer is greater than about half of the total resistance, the gain adjustment should be fairly fine. MATH CHALLENGE: For a gain of 83.4, how much of the potentiometer, in kilohms, would act as R_i ?

Your response	Correct response
1	1.00719±5E-4

Auto graded Grade: 0/1.0 ✖

$k\Omega$. Suffice it to say that the fixed resistor and potentiometer have been chosen to make the gain adjustment relatively straightforward.

- **Calibration:** Since the gain shoots up wildly when the potentiometer is close to the "left" side, it's likely that the output will be "nailed to the rails" if the potentiometer is set towards that end. The best approach to calibration would be to set the potentiometer to the other end (low gain) and adjust the gain up.
- An unfortunate side effect of adjusting the gain of the final stage of the circuit is that it affects both the signal span and the offset voltage, and for a full calibration this would mean a number of iterations until both the gain and the offset were sufficiently accurate. However, we can use this circuit feature to our advantage in a rough calibration of the system, which will be good enough for this exercise.
- What offset is required to make the range of differential input signals map to the desired output range of -7.5 V to +7.5 V?

Your response	Correct response
-3.51	3.526±0.01

Auto graded Grade: 0/1.0 ✖

V

- Given that the offset into the INA114 will be amplified by the U3C circuit, what offset needs to be injected into the INA114, in millivolts?

Your response	Correct response
41.2	42.3±0.3

Auto graded Grade: 0/1.0 ✖

mV

- Since the offset into the INA114 is generated from the REF02, what is the overall gain required of U3A and U3B combined? (This will be tiny!)

Your response	Correct response
0.00842	0.00846±1E-4

Auto graded Grade: 1/1.0 ✔

- As you know, the gain of a non-inverting amplifier cannot be less than one; but we need a tiny positive offset. In other activities, we've achieved this using a three-resistor variable voltage divider and a buffer. In this case, we've got a quad op amp on board, so we might as well use up two of the available amps as two cascaded inverting amplifiers. What is the gain of the U3A amplifier? (Include both polarity and magnitude.)

Your response	Correct response
-0.1	-0.1

Auto graded Grade: 1/1.0 ✓

. How much gain is required of the U3B amplifier (both polarity and magnitude)?

Your response	Correct response
-0.082	-0.0846±0.001

Auto graded Grade: 0/1.0 ✗

- Notice the unusual arrangement of R3 and R4. Again, the potentiometer is split into two parts: one part is R_f , the other part combines with R3 to become R_i . When the R4 sweeper is all the way to the left (i.e. $R_f = R4$), what is the gain of U3B (both polarity and magnitude)?

Your response	Correct response
-0.227	-0.2273±0.003

Auto graded Grade: 1/1.0 ✓

. When the R4 sweeper is all the way to the right, what can we say about the gain of U3B?

Your response	Correct response
it is zero	it is zero

Auto graded Grade: 1/1.0 ✓

. MATH CHALLENGE: What portion of R4 needs to be used as R_i (i.e. to the left of the sweeper), in kilohms, to set the gain of U3B to -0.0846?

Your response	Correct response
2.89	2.12±0.02

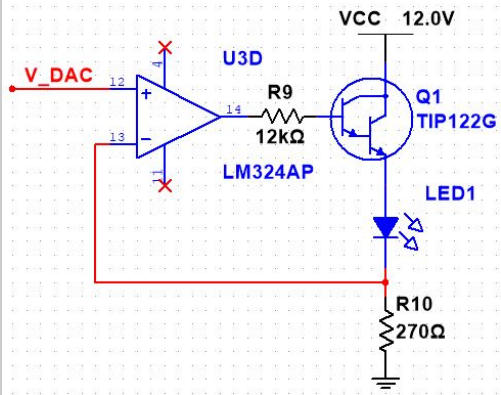
Auto graded Grade: 0/1.0 ✗

kΩ

Control Circuit

For this system, the output is simply an indicator of how much the temperature differs from a set-point. More specifically, an LED will ramp up and down in brightness at a rate proportional to the difference between the temperature and the set-point.

In order to vary the brightness of the LED, an analog signal will be generated by the LabVIEW VI through an analog output (DAC) from the NI-USB6221 digitizer. The voltage will fairly slowly ramp up and down between 0.0 V and 5.0 V. The following Voltage to Current converter will vary the brightness of the LED.



Notes:

- The op amp in this circuit is the fourth op amp in the quad LM324.
- The TIP122 provides the current for the LED through R10.
- R9 is present to protect the op amp in the event of a catastrophic failure of the TIP122.
- Assume a forward voltage for the LED of 1.8 V.
- Click this link to access the [TIP122 data sheet \(https://www.onsemi.com/pdf/datasheet/tip120-d.pdf\)](https://www.onsemi.com/pdf/datasheet/tip120-d.pdf). From this, use the minimum β and the maximum V_{BE} .

12. What voltage should appear at the top of R10?

Your response	Correct response
	V_DAC

Auto graded Grade: 0/1.0

13. If the voltage across R10 is 5.00 V, what is I_E for the TIP122, in milliamps?

Your response	Correct response
18.5	18.52±0.03

Auto graded Grade: 1/1.0

mA

- For the conditions above, what is the expected Base current for the TIP122, in microamps?

Your response	Correct response
18.5	18.52±0.03

Auto graded Grade: 1/1.0

μA

- Again, for the same conditions, what voltage would be expected at the output of U3D?

Your response	Correct response
8	9.52±0.03

Auto graded **Grade: 0/1.0**

$$\overline{v}_i$$

- From the [LM324 data sheet](https://www.ti.com/lit/ds/symlink/lm324.pdf) (<https://www.ti.com/lit/ds/symlink/lm324.pdf>?ts=1669793627455&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FLM324), the LM324 is capable of driving the output to within 1.75 V of the rails (*see if you can verify this statement*). TRUE/FALSE: This indicates that the choice of 12 k Ω is suitable as the Base resistor in this application:

Your response	Correct response
True	True

Auto graded **Grade: 1/1.0**

-

- When V_{DAC} is 1.0 V, what is the expected current through the LED, in milliamps?

Your response	Correct response
3.7	3.7±0.1

Auto graded **Grade: 1/1.0**

mA

You have now completed the In-Class question. Do not Submit yet, or you will be prevented from accessing the rest of the project!

Please either "Save and Close" or move on in Mobius to the next question to ensure that access to this question is blocked for administrative purposes.

✘ Total grade: $1.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 + 1.0 \times 1/24 + 0.0 \times 1/24 + 1.0 \times 1/24 = 4\% + 4\% + 4\% + 4\% + 0\% + 4\% + 4\% + 0\% + 4\% + 0\% + 0\% + 0\% + 4\% + 4\% + 0\% + 4\% + 4\% + 0\% + 4\% + 4\% + 0\% + 4\% + 4\%$